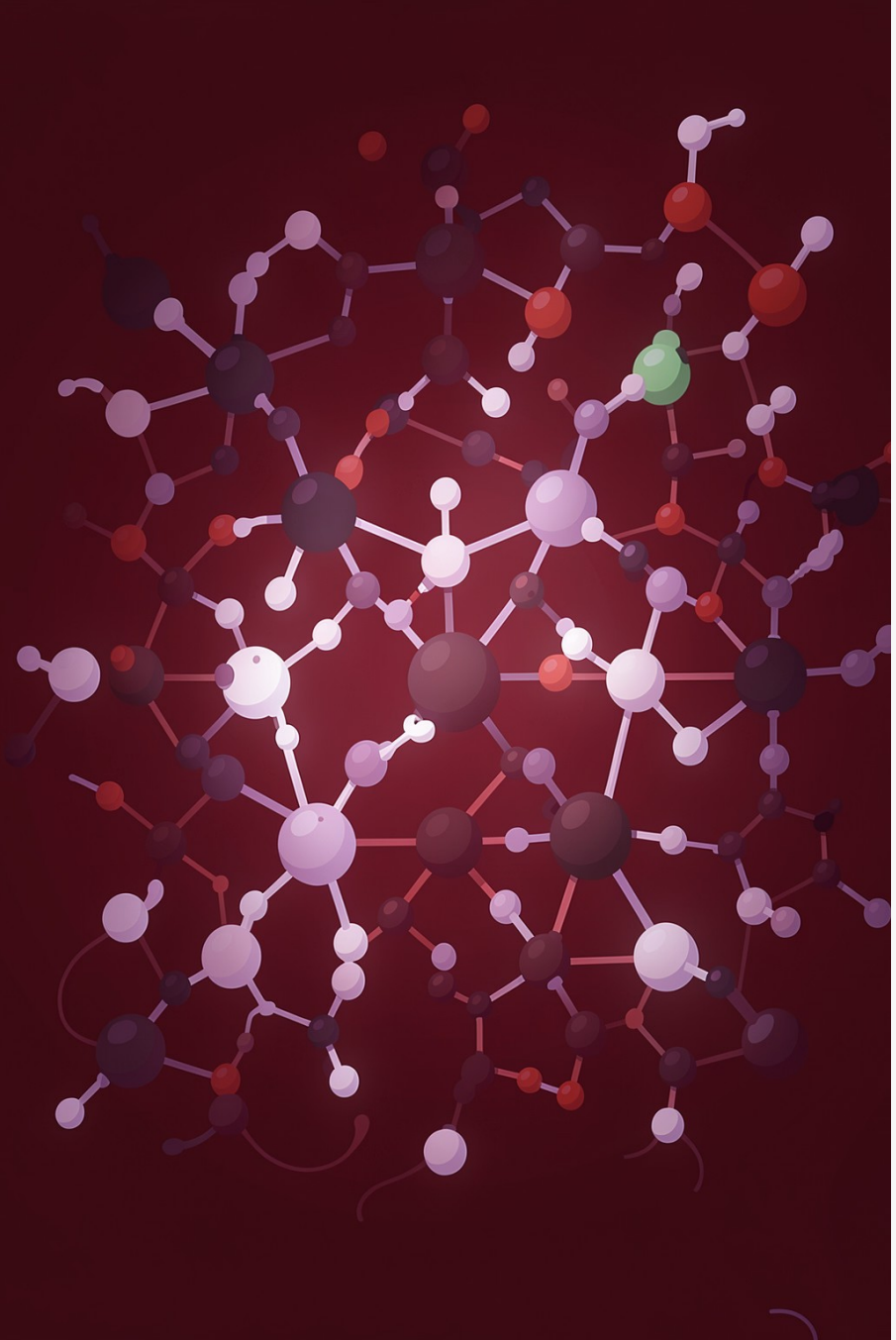


Frank's Model

The Origin of Biological Homochirality: Frank's Autocatalytic Model

How can a symmetric world produce asymmetric life?



by Saswata Chatterjee

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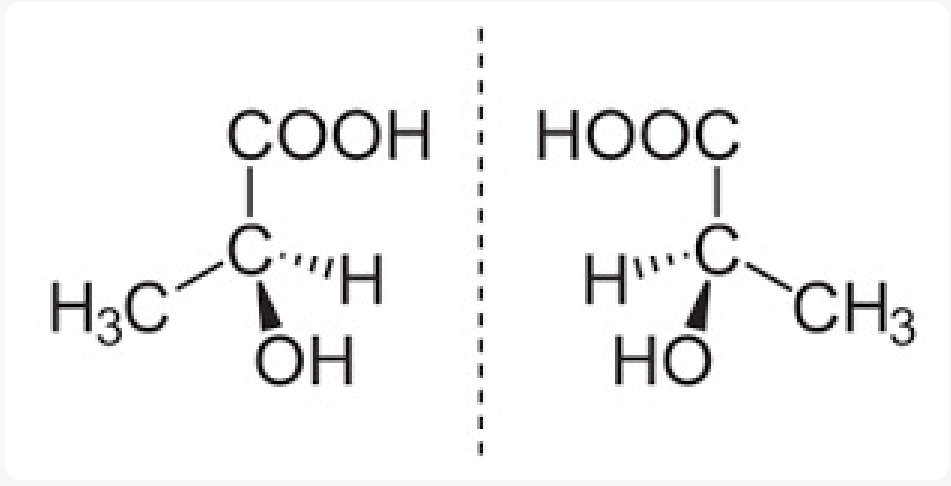
Chirality, Enantiomers and Racemic Mixtures

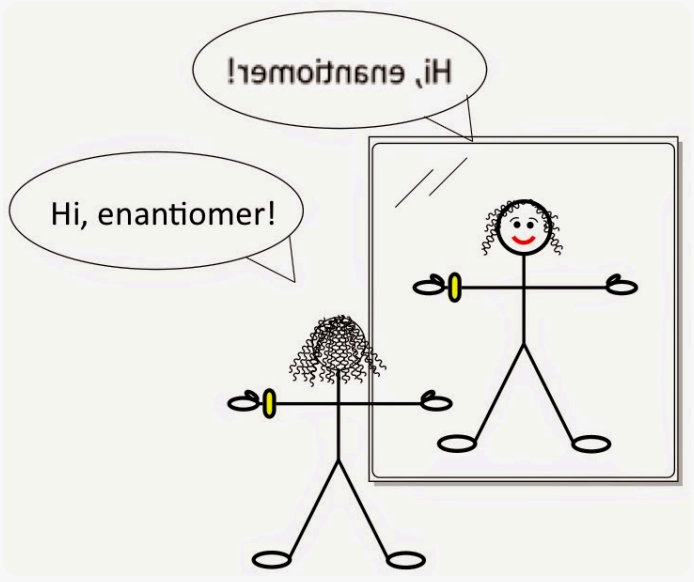
Chirality

- A chiral molecule **cannot** be superimposed on its mirror image.
- Its mirror-image forms are called **enantiomers**.
    - xThey are often labeled **L** (left-handed) and **D** (right-handed).
- **Amino Acids:** Nearly exclusively **L-chiral**.
- **Sugars:** Nearly exclusively **D-chiral**.

Racemic Mixture

- Without a chiral influence, reactions usually form both enantiomers equally.
- A **50:50 mix** of L and D enantiomers is a **racemic mixture**.
- It has **no overall chirality**.





These two molecules are **Enantiomers** as they are **non super-imposible** and **mirror images**

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Frank's Original Model (1953)

Frank proposed the first theoretical model explaining how **spontaneous symmetry breaking** can produce homochirality through autocatalysis and mutual antagonism.

Reaction Scheme

Autocatalytic Production for D

$$A + D \xrightarrow{k_a} 2D$$

Autocatalytic Production for L

$$A + L \xrightarrow{k_a} 2L$$

Mutual Antagonism

$$D + L \xrightarrow{k_i} \varnothing$$

Key Assumptions

- The achiral precursor concentration is held constant: $[A] = a = \text{constant}$
- The system is treated as an **open system**, with fresh A supplied externally.
- Opposite enantiomers inhibit each other through **mutual antagonism**, leading to amplification of an initial chiral imbalance.

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Limitation of Frank's Original Model

Although Frank's model successfully demonstrates **spontaneous symmetry breaking**, it does **not** possess a finite steady state.

Problem Statement

The achiral precursor is assumed to remain constant:

$$[A] = a = \text{constant}$$

This means fresh A is continuously supplied, allowing autocatalysis to proceed indefinitely.

Asymptotic Behavior

As the system approaches homochirality:

$$L \to 0$$

the dominant enantiomer satisfies:

$$\frac{dD}{dt} \approx k_a a D.$$

Rate Equations

The rate equations are:

$$\frac{dD}{dt} = k_a a D - k_i D L$$

$$\frac{dL}{dt} = k_a a L - k_i D L$$

In this limit, the mutual antagonism term becomes negligible, and the dominant species grows according to simple exponential dynamics.

Key Result

$$D(t) = D_0 e^{k_a a t}$$

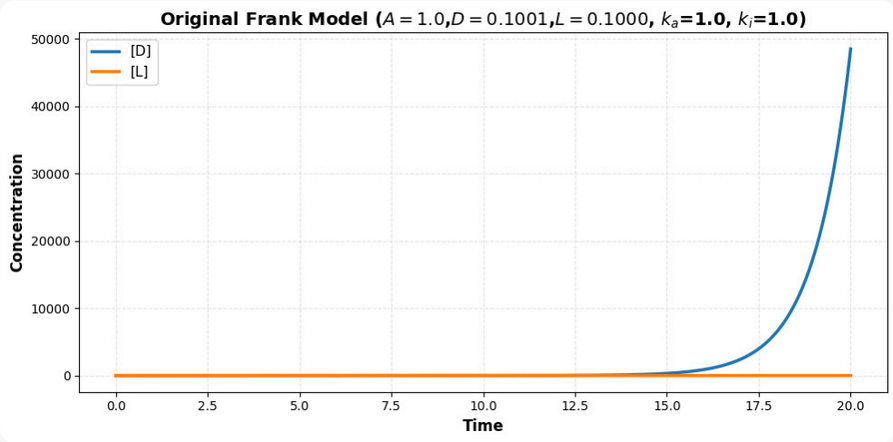
This growth is unbounded. The model therefore cannot reach a finite equilibrium concentration.

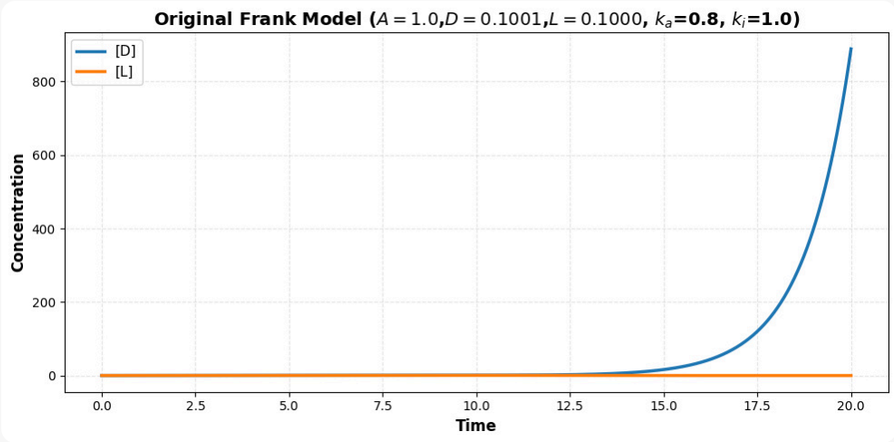
Consequence

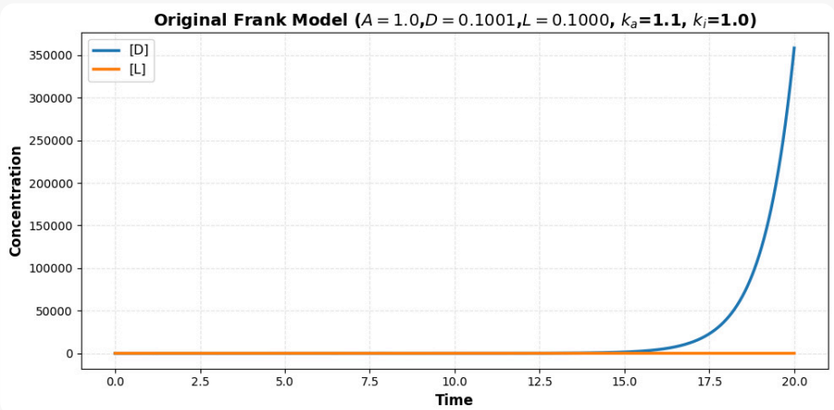
- Continuous external supply of A sustains autocatalysis indefinitely.
- The dominant enantiomer never reaches a finite equilibrium concentration.
- The model therefore has **no well-defined steady state**, motivating later molecule-conserving modifications.

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Some key results:







Conclusion:

- The original Frank model is highly unrealistic as there is no fixed steady state for the system and there is an open system an a potentially infinite supply for A molecules.
- We look at a modified reaction scheme for the Frank Model.

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Why Not Simply Let A Vary?

Suppose Frank's original reactions are retained but the assumption $[A] = \text{constant}$ is removed.

- The autocatalytic reactions consume the achiral precursor:

$$A + D \rightarrow 2D, \qquad A + L \rightarrow 2L.$$

- However, the inhibition reaction $D + L \rightarrow \varnothing$ permanently removes molecules from the system. There will be a point of time the concentration of A and L will both be 0 as A is never replinished.
- The total number of molecules is therefore **not conserved**:

$$\frac{d}{dt}(A + D + L) = -2k_i DL < 0$$

- If one attempts to formulate Frank's original reaction scheme as a closed system by allowing A to vary, the reaction violates conservation of mass because molecules disappear from the system..

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Frank's Model of Homochirality (Modified)

Frank proposed that chiral molecules are produced autocatalytically from an achiral precursor A.

Autocatalytic Production

$$A + D \xrightarrow{k_a} 2D$$

$$A + L \xrightarrow{k_a} 2L$$

where:

- A = achiral precursor
- D = D-enantiomer
- L = L-enantiomer
- k_a = autocatalytic rate constant

Chiral Inhibition

$$D + L \xrightarrow{k_i} 2A$$

where:

- k_i = chiral inhibition rate constant

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Modifications introduced in Frank Model (Jafarpour et al.)

Jafarpour *et al.* modified Frank's original reaction scheme to obtain a **well-defined steady state** while preserving the symmetry-breaking mechanism.

Original Model

- The achiral precursor concentration is held constant.
- Chiral inhibition is represented by

$$D + L \rightarrow \varnothing$$

- This creates an **open system** with no conservation of the total number of molecules.

Modified Model

- The inhibition reaction is replaced by

$$D + L \rightarrow 2A$$

- Opposite enantiomers are recycled back into the achiral precursor.
- The total number of molecules is conserved:

$$[A] + [D] + [L] = N$$

- This produces a finite, well-defined steady state and simplifies the mathematical analysis.

Key difference: the modified model closes the system by recycling D and L into A, enabling a stable steady state.

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Chiral Order Parameter

The state of the system is described by the chiral order parameter

$$\omega = \frac{[D] - [L]}{[D] + [L]}$$

Interpretation

- $\omega = 0 \rightarrow$ Racemic mixture
- $\omega = +1 \rightarrow$ Pure D state
- $\omega = -1 \rightarrow$ Pure L state

Objective

Explain how a system starting near

$$\omega \approx 0$$

can evolve toward

$$|\omega| = 1$$

through spontaneous symmetry breaking.

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Mean-Field Dynamics of Frank's Model

The reactions in Frank's model can be translated into a set of deterministic **mean-field equations** using mass-action kinetics.

Mean-Field Equations

$$\frac{d[A]}{dt} = 2k_i[D][L] - k_a[A]\big([D]+[L]\big)$$

$$\frac{d[D]}{dt} = k_a[A][D] - k_i[D][L]$$

$$\frac{d[L]}{dt} = k_a[A][L] - k_i[D][L]$$

Parameters

- $[A]$: concentration of the achiral precursor
- $[D]$: concentration of the D-enantiomer
- $[L]$: concentration of the L-enantiomer
- k_a : autocatalytic rate constant
- k_i : chiral inhibition rate constant

Evolution of the Order Parameter

Using the mean-field equations, the evolution of ω is

$$\frac{d\omega}{dt} = \frac{1}{2}k_i([D]+[L])\,\omega\,(1-\omega^2)$$

Key Result

The factor

$$\omega(1-\omega^2)$$

produces three fixed points:

$$\omega = -1, \quad 0, \quad +1$$

representing two homochiral states and one racemic state. This three-fixed-point structure is the mathematical basis of Frank's spontaneous symmetry-breaking mechanism.

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Stability Analysis

Understanding whether an equilibrium is stable, unstable, or a saddle point helps determine how a dynamical system responds to small perturbations.

Fixed Point

A fixed point is a state where the system no longer evolves.

$$\frac{dD}{dt} = 0, \qquad \frac{dL}{dt} = 0$$

At a fixed point, the concentrations remain constant unless the system is perturbed.

⚠ Nearby trajectories reveal whether the equilibrium attracts, repels, or behaves differently along distinct directions.

Equilibrium Criteria

Stable Equilibrium

Nearby trajectories return to the fixed point after a small perturbation.

Interpretation: the equilibrium attracts nearby states.

Unstable Equilibrium

Nearby trajectories move away from the fixed point after a small perturbation.

Interpretation: the equilibrium repels nearby states.

Key Takeaway

By combining the fixed-point condition with local stability analysis, we can determine whether an equilibrium is stable, unstable, or a saddle point.

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Linear Stability Analysis :

To characterize the stability of the racemic equilibrium, the conservation law

$$A + D + L = N$$

was first used to eliminate the achiral precursor,

$$A = N - D - L$$

reducing the system to two coupled nonlinear ODEs in D and L. The racemic equilibrium was then obtained by imposing:

$$\dot{D} = \dot{L} = 0$$

By analysing the nullclines we can understand the equilibrium points which would be located at their intersection. However it would reveal results which are not important or trivial , thus together with the symmetry condition $D = L$. We can obtain points which form a **Racemic Equilibrium**.

$$\boxed{D^* = L^* = \frac{k_a N}{2k_a + k_i}, \qquad A^* = \frac{k_i N}{2k_a + k_i}.}$$

For the symmetric case $(k_a = k_i)$:

$$\boxed{A^* = D^* = L^* = \frac{N}{3}.}$$

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Jacobian and Eigenvalue analysis:

The reduced system was linearized about the racemic equilibrium by constructing the Jacobian matrix for these equations:

$$f(D,L) = k_a D(N - D - L) - k_i DL$$

$$g(D,L) = k_a L(N - D - L) - k_i DL$$

The Jacobian is:

$$J = \begin{pmatrix} \dfrac{\partial f}{\partial D} & \dfrac{\partial f}{\partial L} \\ \dfrac{\partial g}{\partial D} & \dfrac{\partial g}{\partial L} \end{pmatrix}$$

Evaluating at the racemic equilibrium gives:

$$J = -\frac{k_a N}{2k_a + k_i} \begin{pmatrix} k_a & k_a + k_i \\ k_a + k_i & k_a \end{pmatrix}$$

The corresponding eigenvalues are:

$$\boxed{\lambda_1 = -k_a N, \qquad \lambda_2 = \frac{k_a k_i N}{2k_a + k_i}}$$

with the eigen vectors:

$$\boxed{\mathbf{v}_1 = (1,1)^T, \qquad \mathbf{v}_2 = (1,-1)^T}$$

Interpretation

- Since k_a, k_i, N are all positive quantities inherently it is fair to say that one of my eigven values is negative while the other is positive.
- Under these conditions all implications lead to my equilibrium point being a Saddle Point as $\lambda 1 < 0, \lambda 2 > 0$. This implies our equilibrium is linearly unstable with perturbations growing along one directions and decaying along the other.
- The racemic equilibrium lies on the symmetry line $\boxed{D = L}$

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Assumptions for Phase Space Analysis

To simplify the qualitative analysis of the modified Frank model, the following assumptions are made:

> For analytical tractability, we impose the symmetry condition $D = L$to isolate the racemic branch of equilibria, rather than solving the full nullcline system for arbitrary $k_a \neq k_i$

The total concentration is normalized

$$A + D + L = 1$$

which is equivalent to scaling the conserved quantity N by the total number of molecules- equal reaction rates are assumed,

$$k_a = k_i = 1$$

to obtain a **symmetric system** where autocatalysis and chiral inhibition are equally weighted. This normalization does not alter the qualitative dynamics or stability of the system, but simplifies the analytical treatment.

Under these assumptions,

The nullcline equations obtained are:

$$\dot{D} = D(1 - D - 2L) = 0$$

$$\dot{L} = L(1 - 2D - L) = 0$$

Solving these, we get 4 points of intersection:

$$\boxed{\left\{(0,0),\ (1,0),\ (0,1),\ \left(\frac{1}{3}, \frac{1}{3}\right)\right\}}$$

The second point which can also be obtained by just computing the parameters in the solved D and L value before is:

$$\boxed{(D^*, L^*) = \left(\frac{1}{3}, \frac{1}{3}\right).}$$

As discussed earlier this is our **Saddle Point.**

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Jacobian Analysis at all these points:

For the normalized modified Frank model,

$$A + D + L = 1, \qquad k_a = k_i = 1$$

The Jacobian matrix is:

$$J(D, L) = \begin{pmatrix} 1 - 2D - 2L & -2D \\ -2L & 1 - 2D - 2L \end{pmatrix}$$

Now we can individually discuss the stability of all of the 4 points which were previously obtained.

---

1. Achiral Equilibrium

$$(D, L) = (0, 0)$$

Jacobian

$$J(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = \lambda_2 = 1$$

Eigenvectors

$$\mathbf{v}_1 = (1, 0)\,, \qquad \mathbf{v}_2 = (0, 1)\,.$$

Stability

Both eigenvalues are positive.

Unstable Source

2. Racemic Equilibrium

$$(D, L) = \left(\frac{1}{3}, \frac{1}{3}\right)$$

Jacobian

$$J = \begin{pmatrix} -\frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & -\frac{1}{3} \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = -1, \qquad \lambda_2 = \frac{1}{3}$$

Eigenvectors

$$\mathbf{v}_1 = (1, 1)\,, \mathbf{v}_2 = (1, -1)$$

Stability:

Saddle Point

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3. D-Homochiral Equilibrium

$$(D, L) = (1, 0)$$

Jacobian

$$J = \begin{pmatrix} -1 & -2 \\ 0 & -1 \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = \lambda_2 = -1$$

Eigenvector

$$\mathbf{v} = (1, 0)$$

The Jacobian is **defective**, possessing only one linearly independent eigenvector.

Stability: Stable Improper Node

4. L-Homochiral Equilibrium

$$(D, L) = (0, 1)$$

Jacobian

$$J = \begin{pmatrix} -1 & 0 \\ -2 & -1 \end{pmatrix}$$

Eigenvalues

$$\lambda_1 = \lambda_2 = -1$$

Eigenvector

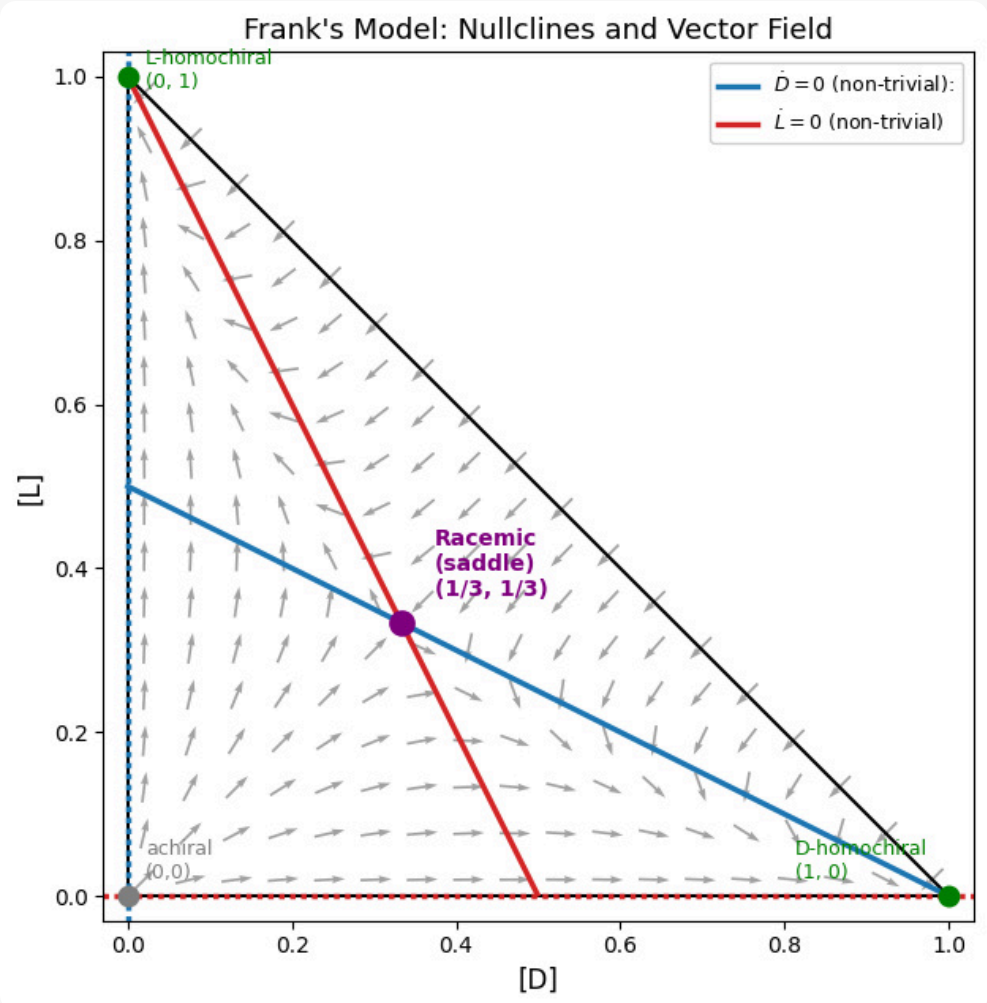
$$\mathbf{v} = (0, 1)$$

The Jacobian is **defective**, possessing only one linearly independent eigenvector.

Stability: Stable Improper Node

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Analysis of the Phase Space:



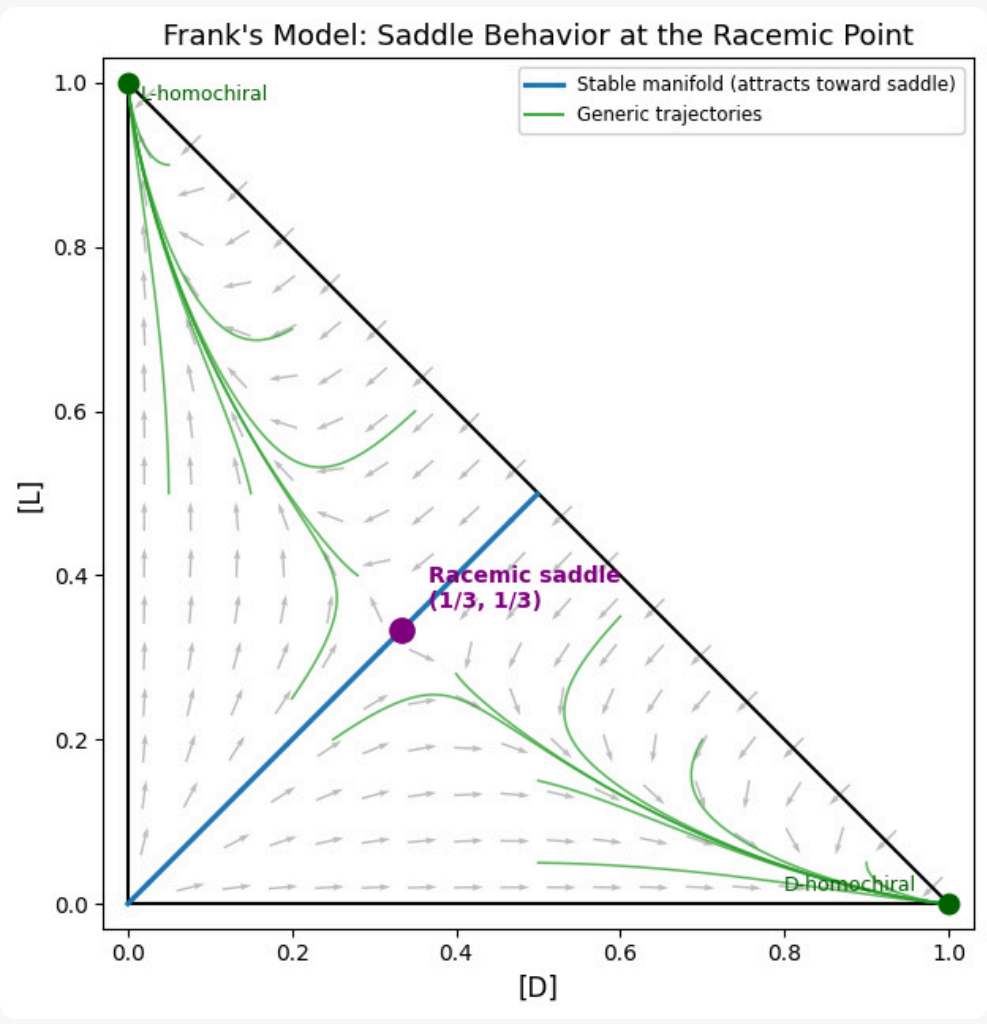
The figure along side is a vector field with the D and L nullcline, there are some interesting inferences we can draw from it:

- The Jacobian eigenanalysis predicted the racemic equilibrium to be a saddle point:

$$\left(\frac{1}{3}, \frac{1}{3}\right)$$

this perfrectly follows our calculation.

- The vector field gives us a great picture which allows us to visualise the Saddle point at our racemic equilibrium.



This figure is specifically illustrating the **saddle nature** of the racemic equilibrium:

- This behaviour is consistent with the analytical eigenvalue analysis
- The stable manifold coincides with the symmetry line

$$D = L$$

corresponding to the eigenvector

$$v_1 = (1, 1)^T$$

- Perturbations perpendicular to this manifold correspond to the unstable eigenvector

$$v_2 = (1, -1)^T$$

which amplifies an initially small enantiomeric imbalance and drives spontaneous symmetry breaking.

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Relating Phase Space to the Order Parameter

Although the stability analysis was performed in the (D, L) phase space, it is entirely consistent with the order parameter ω.

We had previously stated our enantiomeric excess order parameter as:

$$\omega = \frac{D - L}{D + L}$$

The symmetry line $D = L$ corresponds to $\boxed{\omega = 0}$

Two key observations which could be made from our previous calculations are:

1. The saddle behaviour observed at

$$\left(\frac{1}{3}, \frac{1}{3}\right)$$

shows that infinitesimal perturbations away from $D = L$ grow with time, causing

$$\omega = 0$$

to become unstable.

2. As trajectories evolve toward either homochiral equilibrium,

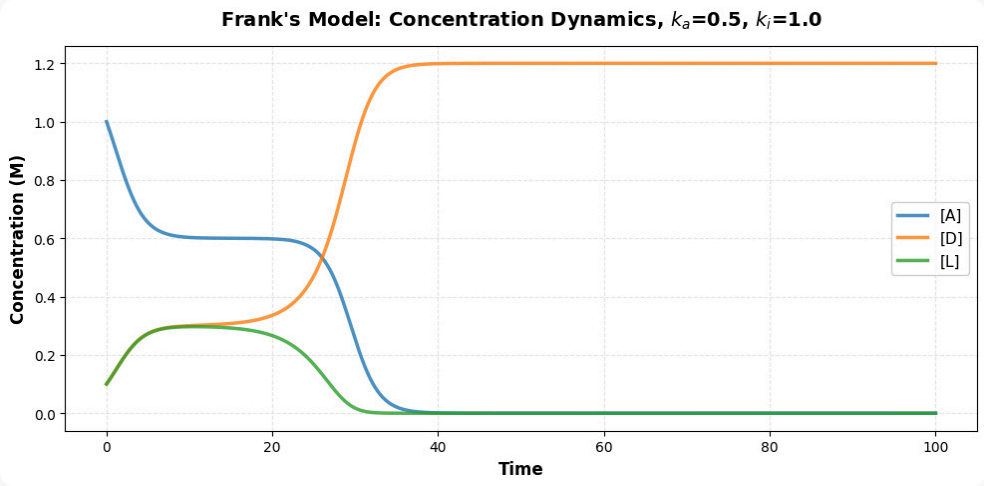
$$\omega \to \pm 1$$

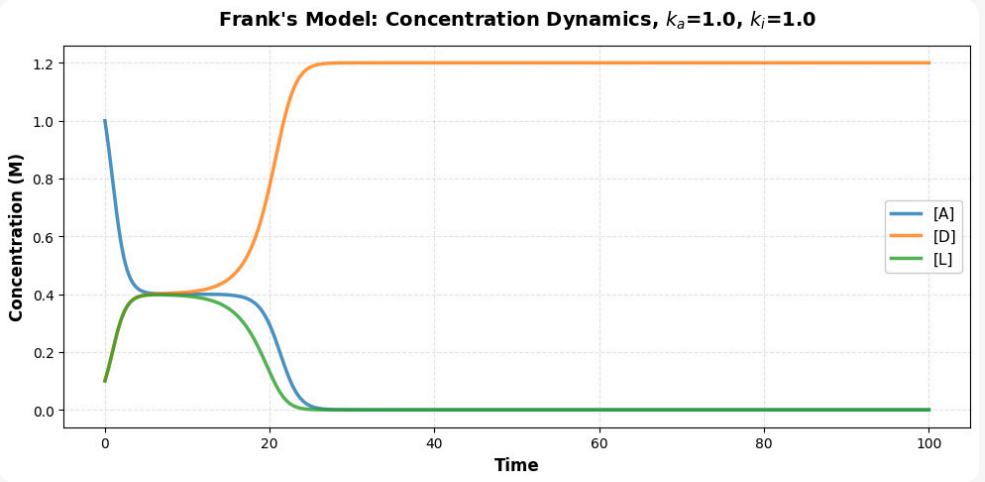
confirming the stable fixed points predicted by the analytical solution of the order parameter equation.

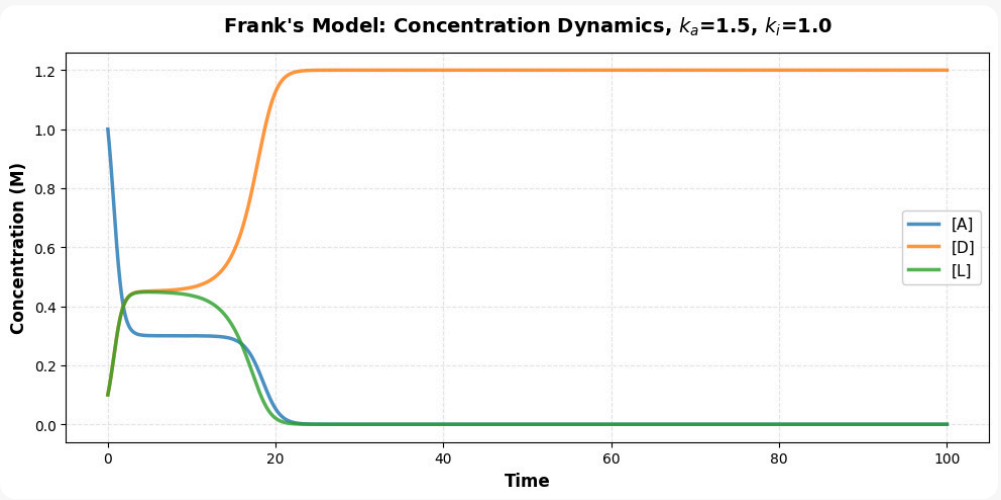
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Deterministic Results

1. $[A_0]$=1.0 , $[D_0]$=0.1001 , $[L_0]$=0.1000, these are the starting concentrations of each speices:







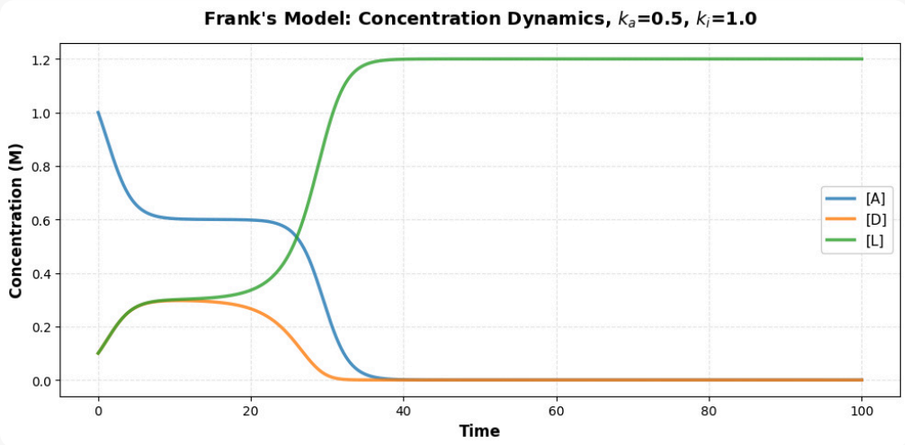
Conclusion

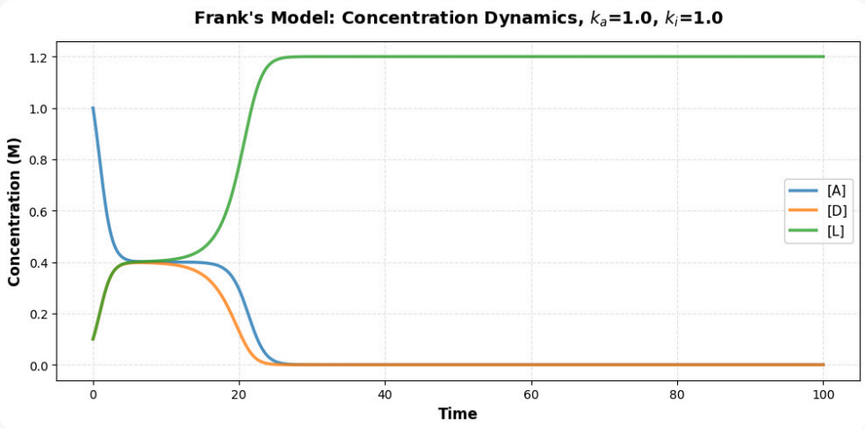
- Even a very small initial dominance is enough for the system to amplify it over time to a state of homochirality
- The k_a: k_i ratio is used to determine how early the leading species take dominance however it cannot change the final outcome

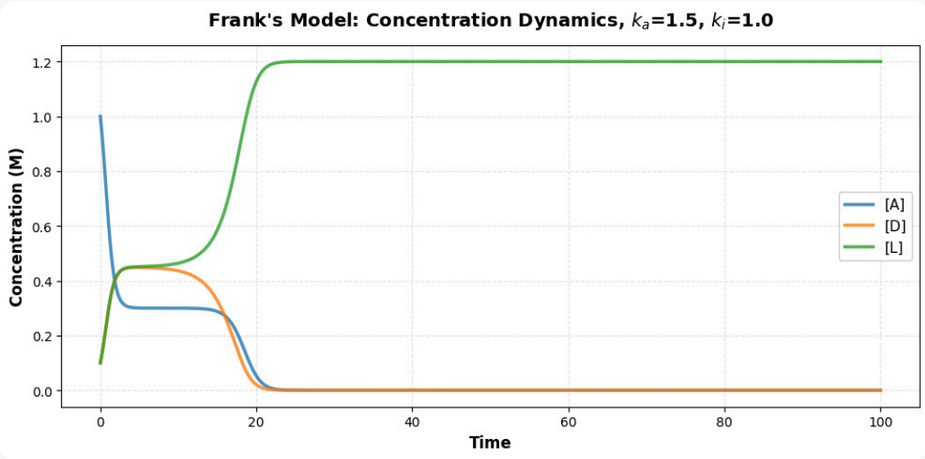
⚠ **NOTE:** The the system drives toward homochirality but **mathematically never reaches it in finite time**. The minority enantiomer decays exponentially toward zero but asymptotically — it takes infinite time to reach exactly zero. So deterministically you get *arbitrarily close* to homochiral but never 100%

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2.$[A_0]$=1.0 , $[D_0]$=0.1000, $[L_0]$=0.1001, these are the starting concentrations of each speices:





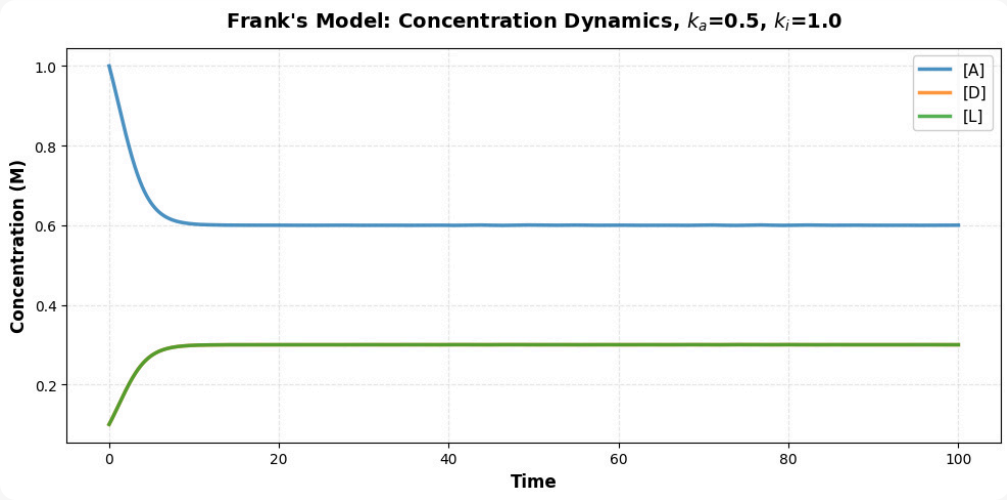


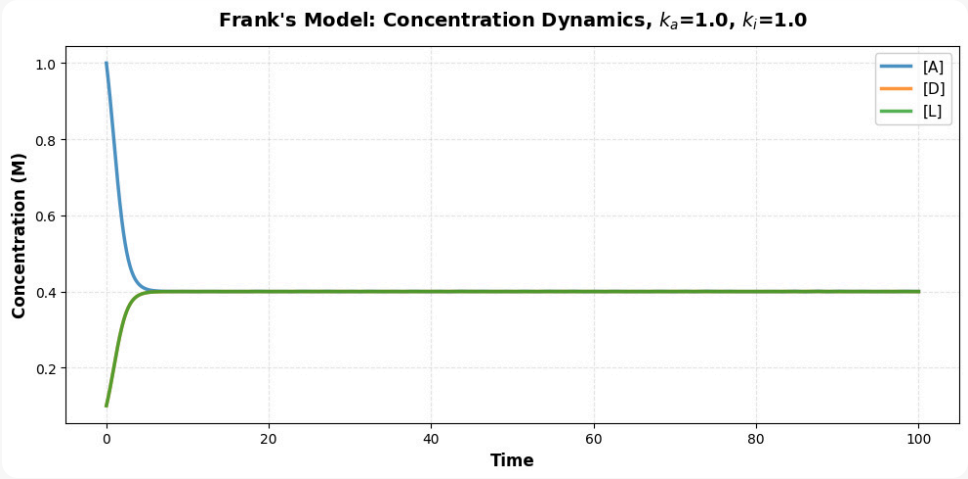
Conclusion:

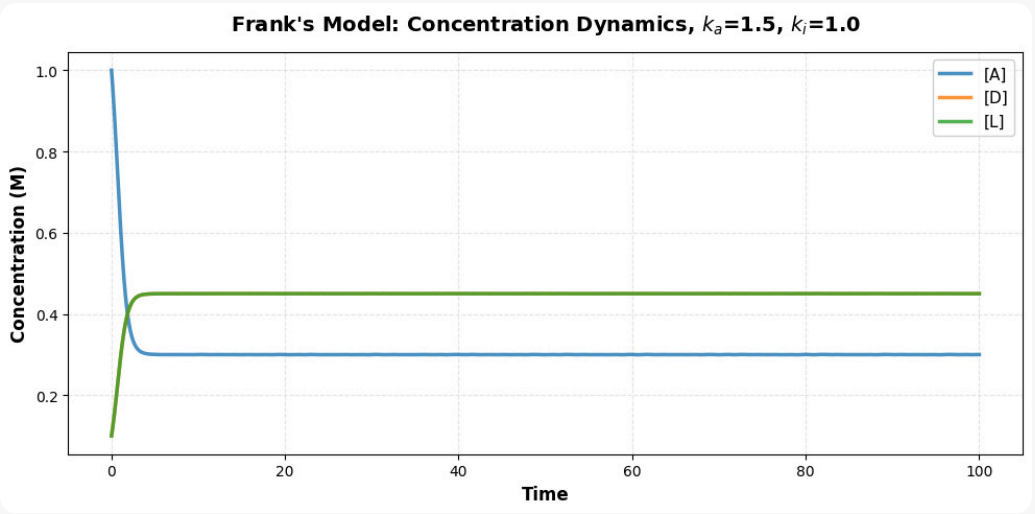
- As expected the L specie at the beginning had a slight edge and it gets amplified to a point of homochirality at the end.
- The k_a: k_i ratio is again only used to determine how fast the species take dominance and not which is the homochiral spieces.

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3.$[A_0]$=1.0 , $[D_0]$=0.1000 , $[L_0]$=0.1000, these are the starting concentrations of each speices:







Conclusion:

- The initial species being a racemic mixture yield a racemic mixture as there is no perturbation in the system to amplify it.
- If we consider an initial constraint that

$$A + D + l = N$$

At equilibrium, the production and inhibition rates of both enantiomers must balance. Since $k_a = k_i$, the equilibrium conditions reduce to

$$A = D = L = N/3$$

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Gillespie Simulation

Since the deterministic model of Frank's Model failed to break the homochirality of the species we wish to examine the stochastic behaviour of our system with the help of Gillespie Simulation. We individually run singular trajectories and analyse something really interesting :

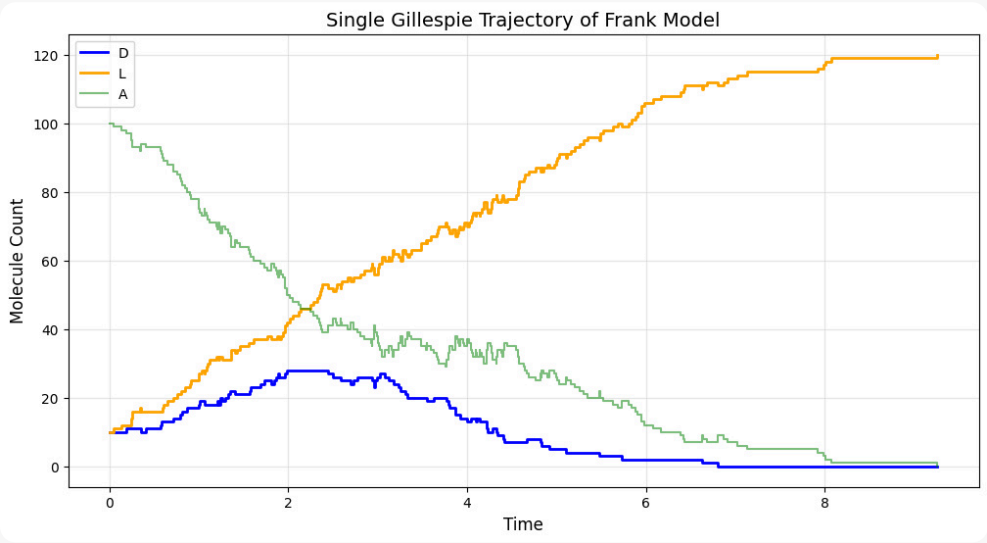
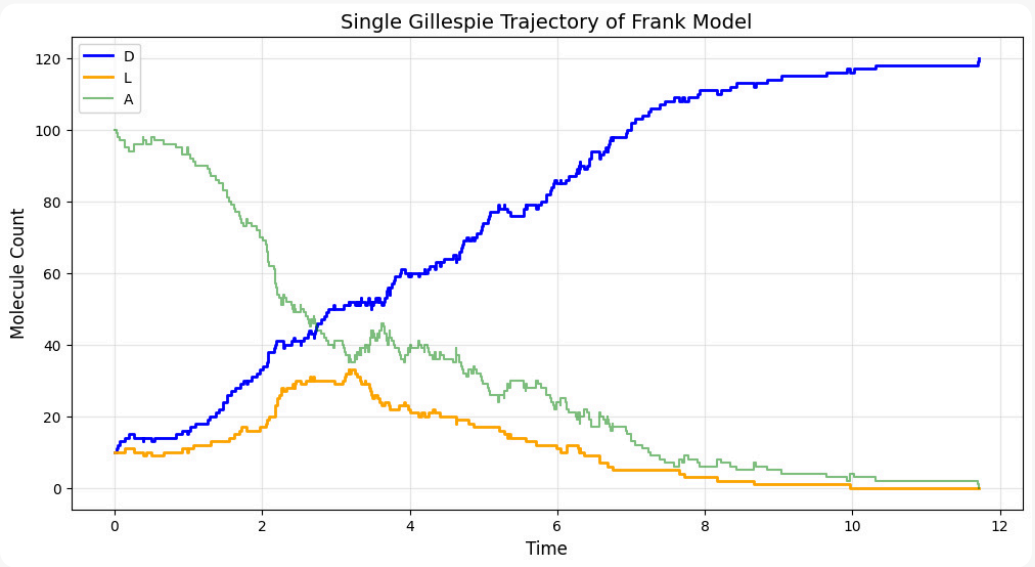


Diagram details:

- k_a = k_i =1.0 (highly inconsequential in determining the dominant species here which matches with the deterministic model).
- A_0 =100, D_0 =10, L_0 =10
- Both trajectories have the same parameter with absolutely no starting bias however L isomer dominates after a short time.



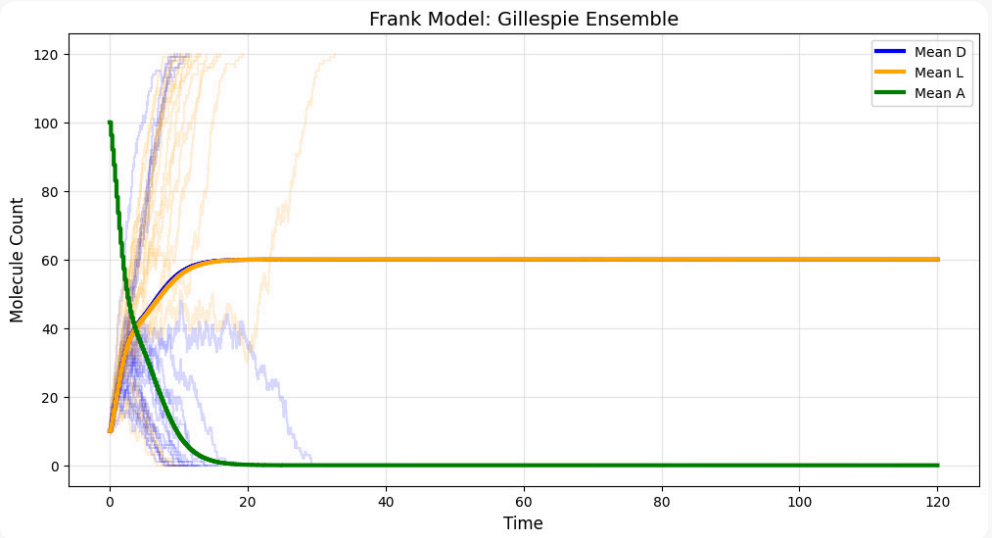
Observation:

This is again an identical simulation which results in a completely different outcome where D is the dominant speices. Running numerous simulations yeilded similar results where the winner was unpredictable.

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Gillespie Simulation over numerous trajectories

The parameters used for these trajectories are k_a = k_i =1.0, A_0 =100, D_0 = L_0 =10, same as the previous gillespie simulations which help us keep the basis of comparison consistent accross these simulations. Furthermore we have simulated 10,000 trajectories.



The faint trajectories are the individual simulations and the bolder ones are the trajcetories representing the Ensemble Mean.

These results are perfectly in line with what was expected, the mean trajectories have essentially averaged out the stochastic behaviour of the singular trajectories however if we do consider the singular trajectories we can clearly see even with a system starting out from a state of Racemic mixture we eventually do obtain **Homochirality**.

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Some points to address regarding the Modified Frank Model

- The modified Frank model provides a mathematically consistent and molecule-conserving framework for studying **spontaneous symmetry breaking** and **homochirality**.
- However, it remains an **idealized theoretical model** and does not describe the detailed molecular mechanism of any specific chemical reaction.
- For over four decades, no experimental chemical system exhibiting **asymmetric autocatalysis** and **enantiomeric excess (EE) amplification** was known.
- The discovery of the **Soai reaction (1995)** provided the first experimental realization of these phenomena, offering strong experimental support for Frank's theoretical framework.
- Consequently, the modified Frank model is widely used as a **minimal theoretical model**, while the Soai reaction serves as its most significant experimental counterpart. However the issue is this was the only experimental evidence regarding the existance and real world significance of this model.
- Recent ribozyme-mediated RNA replication experiments suggest that **homochirality can emerge even in the absence of explicit chiral inhibition**, motivating the study of alternative Frank-type models beyond the original formulation.
- Jafarpour *et al.* replaced the chiral inhibition reaction and tries to study scenarios where we could potentially arrive at homochirality starting from a point of racemic mixture **and** also what would happen if we had means to reintroduce the lost species back into the medium.

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The modified Jafarpour Model

The modified Frank model consists of autocatalytic production of the two enantiomers together with first-order decay back to the achiral precursor

Reaction Model:

$$A + D \xrightarrow{k_a} 2D$$

$$A + L \xrightarrow{k_a} 2L$$

$$D \xrightarrow{k_d} A$$

$$L \xrightarrow{k_d} A$$

Changes:

The chiral inhibition reaction

$$D + L \xrightarrow{k_i} 2A$$

has been replaced by two decay reactions which is in line to our interest in observing potential homochirality in absence of Chiral Inhibition.

Mean-Field Equations:

$$\frac{dA}{dt} = -k_a A(D + L) + k_d(D + L)$$

$$\frac{dD}{dt} = k_a AD - k_d D$$

$$\frac{dL}{dt} = k_a AL - k_d L$$

The total concentration is conserved,

$$A + D + L = N$$

Again N is the total number of molecules present in the system at any given point of time.

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Failure of this model:

There were several reasons as to why this model was rejected,

- In this model homochirality is lost as the ratio of D and L molecules remain constant over time.

The deterministic mean-field equations for the decay model are

$$\frac{dD}{dt} = k_a AD - k_d D$$

$$\frac{dL}{dt} = k_a AL - k_d L$$

To determine how the ratio of the two enantiomers evolves, define $R = \frac{D}{L}$.

$$\frac{d}{dt}\left(\frac{D}{L}\right) = \frac{L\frac{dD}{dt} - D\frac{dL}{dt}}{L^2}$$

Expanding the above equation:

$$\frac{k_a ADL - k_d DL - k_a ADL + k_d DL}{L^2}$$

$$= 0$$

thus effectively all terms are cancelled out.

This demonstrates that the ratio of the two enantiomers is an invariant of the deterministic dynamics. Consequently, the system cannot amplify an initial chiral imbalance. If the system starts from a racemic state. Unlike the Original Franks model there is no reaction where A and D interrract hence there is no way to really amplify the greater enantiomer at a given point of time and obtain a scenario of Homochirality.

Hence, if initially:

$$D_0 = L_0$$

this means,

$$D(t) = L(t)$$

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- To analyse why this model isnt suitable we need to do a dynamic analysis for this we have divided the study in 3 different cases, however to normalise the reactant quantitiy $A + D + L = N = 1$ has always been maintained in all three different cases:

Case 1:

1. Parameters: $k_a = k_d = 1.0$
2. Reduced Equations:

We substitute

$$A = 1 - D - L$$

into the mean-field equations.

The deterministic equations are.

$$\frac{dD}{dt} = D\left[(1 - D - L) - 1\right]$$

which simplifies to

$$\boxed{\frac{dD}{dt} = -D(D + L)}$$

Similarly,

$$\frac{dL}{dt} = L\left[(1 - D - L) - 1\right]$$

giving

$$\boxed{\frac{dL}{dt} = -L(D + L)}$$

3. Solving the nullcline equations we arrive

$$\frac{dL}{dt} = \frac{dD}{dt} = 0$$

hence, after solving these nullclines, we have only one viable solution:

$$\boxed{(D, L) = (0, 0)}$$

This is the only possible equilibrium point which is physically consistent.

4. Linear stability analysis using Jacobian matrix:

The Jacobian matrix is given by:

$$J(D, L) = \begin{pmatrix} \frac{\partial f}{\partial D} & \frac{\partial f}{\partial L} \\ \frac{\partial g}{\partial D} & \frac{\partial g}{\partial L} \end{pmatrix}$$

Evaluating the partial derivatives

$$J(D, L) = \begin{pmatrix} -(2D + L) & -D \\ -L & -(D + 2L) \end{pmatrix}$$

Jacobian at the Equilibrium:

$$J(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

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5. Eigenvalue Analysis

The characteristic equation is

$$\det(J - \lambda I) = 0$$

which yields

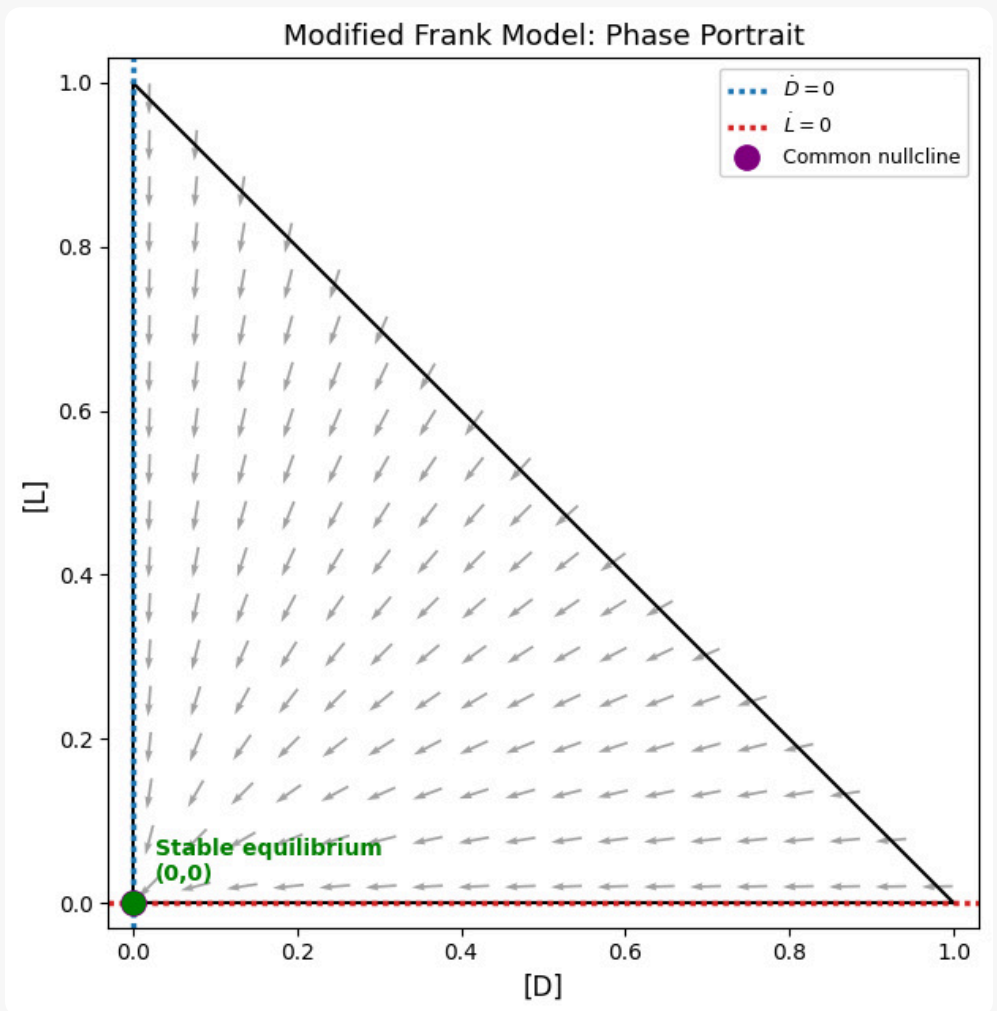
$$\begin{vmatrix} -\lambda & 0 \\ 0 & -\lambda \end{vmatrix} = 0$$

hence the obtained eigen values are:

$$\boxed{\lambda_1 = \lambda_2 = 0}$$

Thus in this scenario the eigen value decomposition and analysis via the jacobian matrix is inconclusive and doesnt yield any significant information.

7. Phase space analysis:



This model can never reach homochirality under the designated conditions as all points would take it to 0.

The reaction has no way of amplifying the more dominant specie and hence the decay catches up and it can never reach a state of homochirality as D and L both just decay.

6. Lyapunov Test:

Consider the Lyapunov function

$$V(D, L) = D + L$$

Positive Definiteness

Since

$$D \geq 0, \qquad L \geq 0$$

it follows that

$$V(D, L) \geq 0$$

with equality only when

$$D = L = 0$$

Hence this function is **positive definite**

Differentiating the Lyapunov function with respect to time,

$$\boxed{\dot{V} = -(D + L)^2}$$

Hence this is clear that this function is **negative definite.**

Thus we can declare that out equilibrium point $(0, 0)$ is **asymptotically stable.**

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Case 2:

1. Parameters: k_a=0.5 k_d=1.0

2 .Reduced Equations

The deterministic equations are:

$$\frac{dD}{dt} = k_a AD - k_d D$$

and

$$\frac{dL}{dt} = k_a AL - k_d L$$

Substituting,

$$k_a = 0.5, \qquad k_d = 1$$

we obtain:

$$\boxed{\frac{dD}{dt} = -\frac{1}{2} D(1 + D + L)}$$

and

$$\boxed{\frac{dL}{dt} = -\frac{1}{2} L(1 + D + L)}$$

3. Nullclines

Setting

$$\frac{dL}{dt} = \frac{dD}{dt} = 0$$

Hence:

$$-\frac{1}{2} L(1 + D + L)$$

$$= -\frac{1}{2} D(1 + D + L) = 0$$

After simplifying these nullcline equations and numerically solving them, we must keep in mind we have to satisfy these two conditions always

$$D \geq 0, \qquad L \geq 0.$$

hence there is only one physically admissible solution

$$D = L = 0$$

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4. Linear Stability Analysis using Jacbobian:

$$J(D,L) = \begin{pmatrix} \frac{\partial f}{\partial D} & \frac{\partial f}{\partial L} \\ \frac{\partial g}{\partial D} & \frac{\partial g}{\partial L} \end{pmatrix}$$

after evaluating the partial derivatives

$$J(D,L) = \begin{pmatrix} -\frac{1}{2}(1+2D+L) & -\frac{1}{2}D \\ -\frac{1}{2}L & -\frac{1}{2}(1+D+2L) \end{pmatrix}$$

The only physically admissible equilibrium is

$$(D,L) = (0,0)$$

Evaluating the Jacobian at this point,

$$J(0,0) = \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

Eigenvalue Analysis

The characteristic equation is

$$\det(J - \lambda I) = 0$$

this yields:

$$\begin{vmatrix} -\frac{1}{2} - \lambda & 0 \\ 0 & -\frac{1}{2} - \lambda \end{vmatrix} = 0$$

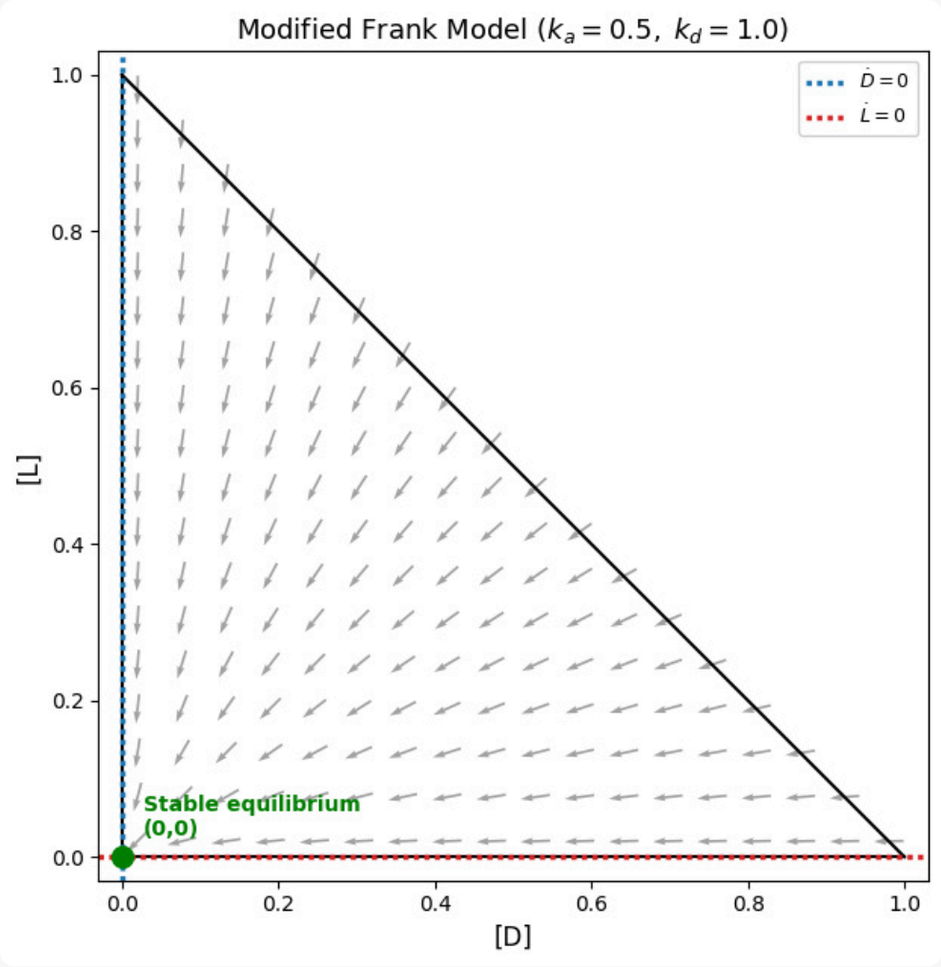
thus our corresponding eigen values are:

$$\boxed{\lambda_1 = \lambda_2 = -\frac{1}{2}.}$$

Since both the eigen values are negative we have a stable equilibrum at $D = L = 0$

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Phase plane analysis



As predicted by our calculations all trajectories will eventaully converge into the point (0,0) as it is the global stable fixed point in this case. This is again another reason we cannot obtain a homochiral system from a racemic mixture.

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Case 3:

1. Parameters : k_a=1.0 ;k_d=0.5
2. Reduced equations:

Substituting the precursor $A = 1 - D - L$ in the original mean field equations we obtain

$$\boxed{\frac{dD}{dt} = D\left(\frac{1}{2} - D - L\right)}$$

and

$$\boxed{\frac{dL}{dt} = L\left(\frac{1}{2} - D - L\right)}$$

3. Nulcline analysis:

Setting

$$\frac{dD}{dt} = \frac{dL}{dt} = 0$$

we obtain the reduced forms as

$$D\left(\frac{1}{2} - D - L\right) = 0$$

and

$$L\left(\frac{1}{2} - D - L\right) = 0$$

4. The equilibrium

By solving the nullclines together we obtain

$$\boxed{(D, L) = (0, 0)}$$

This is the **Achiral Equilibrium point.**

We also obtain an interesting result:

$$\boxed{D + L = \frac{1}{2}, \qquad 0 \leq D \leq \frac{1}{2}.}$$

All points on this line present are corresponding to an equilibrium. Hence we have an entire manifold of equilibrium points here.

$$\boxed{(A, D, L) = \left(\frac{1}{2},\ D,\ \frac{1}{2} - D\right), \qquad 0 \leq D \leq \frac{1}{2}}$$

This is an interesting result we must analyse further.

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5. Jacobian stability analysis:

A. Point: $(D, L) = (0, 0)$

For this scenario the derived Jacobian matrix becomes:

$$J(D, L) =$$

$$\begin{pmatrix} \frac{1}{2} - 2D - L & -D \\ -L & \frac{1}{2} - D - 2L \end{pmatrix}$$

this becomes

$$J(0, 0) = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

Eigenvalue Analysis

by using the characteristic equation:

$$\det(J - \lambda I) = 0$$

we can perform an eigen decomposition and obtain:

$$\begin{vmatrix} \frac{1}{2} - \lambda & 0 \\ 0 & \frac{1}{2} - \lambda \end{vmatrix} = 0$$

Hence

$$\boxed{\lambda_1 = \lambda_2 = \frac{1}{2}}$$

as we can see both our eigen values are positive thus it is an unstable equilibrium and trajecotories diverge away from this position.

B. The line containing equilibria points: $D + L = \frac{1}{2}, \qquad 0 \leq D \leq \frac{1}{2}$

The derived jacobian here becomes:

$$J = \begin{pmatrix} -D & -D \\ D - \frac{1}{2} & D - \frac{1}{2} \end{pmatrix}$$

Here we will not plug any values and try to directly solve the characteristic equation in terms of , only substitution from original jacobian we made was $D = (1/2) - L$

Eigenvalue Analysis

by analysing the characteristic equation we end up with:

$$\det(J - \lambda I) = 0$$

this essentially converts to :

$$J - \lambda I = \begin{pmatrix} -D - \lambda & -D \\ D - \frac{1}{2} & D - \frac{1}{2} - \lambda \end{pmatrix}$$

after solving the determinant we obtain:

$$(-D - \lambda)\left(D - \frac{1}{2} - \lambda\right) + D\left(D - \frac{1}{2}\right)$$

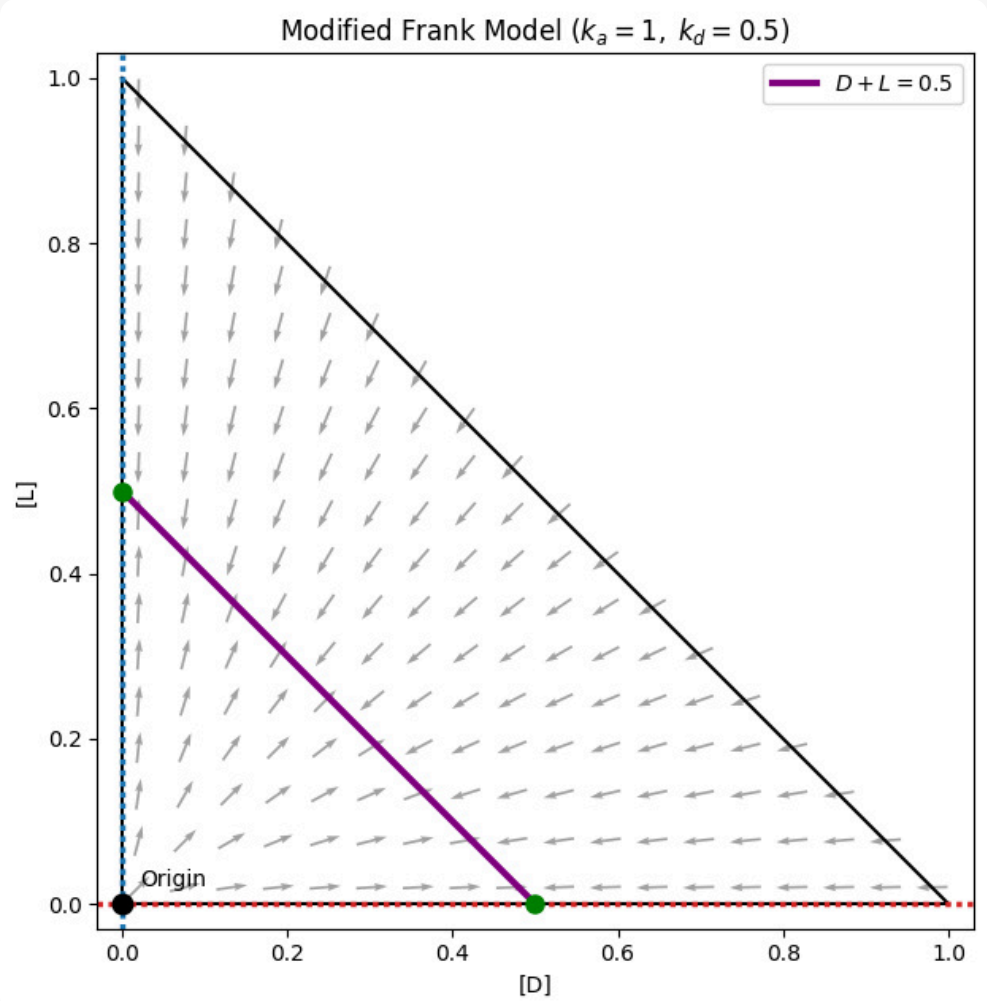
if we solve this we get two eigen values independent of the D term:

$$\boxed{\lambda_1 = 0, \qquad \lambda_2 = -\frac{1}{2}}$$

Now we essentially have an entire line of a normally attracting equilibrium manifold. If we analyse this we have no dynmics for the 0 eigen value and for the negative eigen value if we perturb the trajectory slightly the trajectories return to the line. Hence this is **Stable in a singualr direction.**

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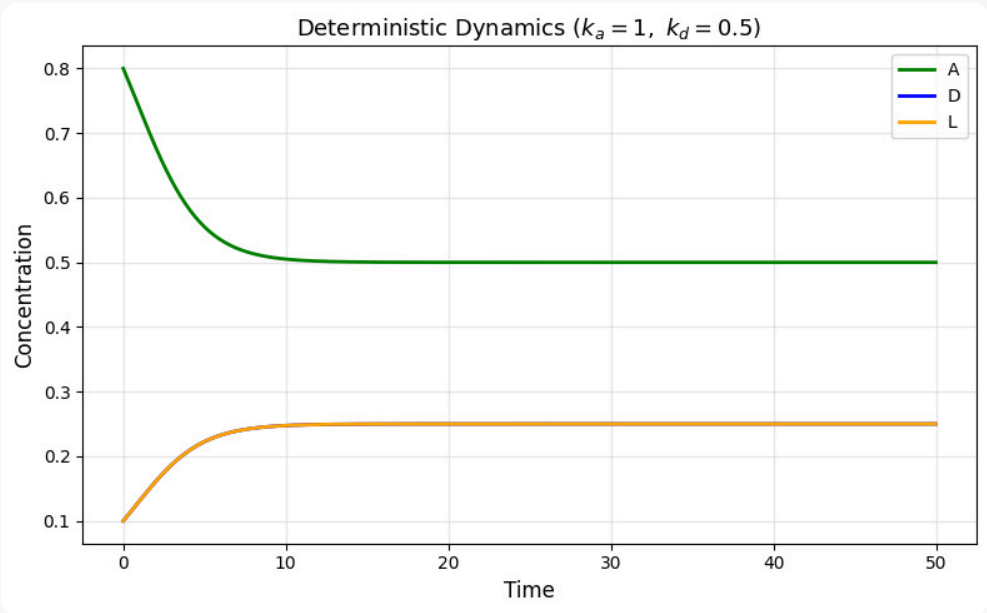
Phase Plane analysis:



This result lines perfectly with our previous studies as we initially discussed that the (D/L)ratio remains constant so depending on where we start our trajectories will always follow the straight line reaching the equilibrium manifold. A similar case is demonstrated in the ODE simulation alongside this.

ODE simulation:

A_0=0.8; $D_0 = L_0$=0.1 (starting conditions)



Since we have the the same position of starting for D and L the trajectories have superimposed however if we do consider that to be true and add them up it infact does like with the fact that L_t= D_t=0.25 at steady state so they do add up and perfectly get attracted to the trajectory

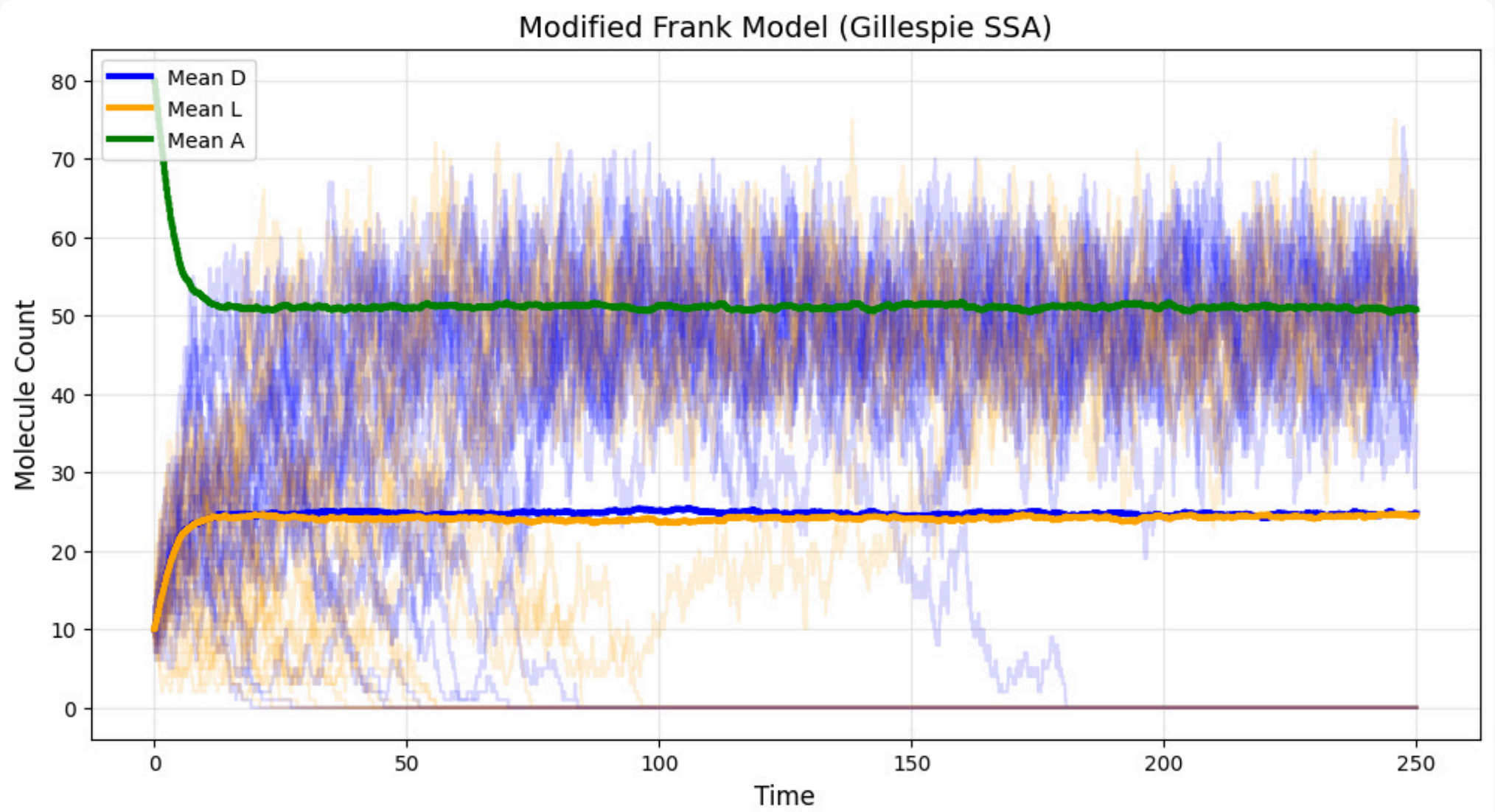
$$A + D = \frac{1}{2}$$

In this particular example since we have toned down the decay rate it might be worth looking into stochastically as to what happens as the trajectories unlike other cases do not always head to 0.

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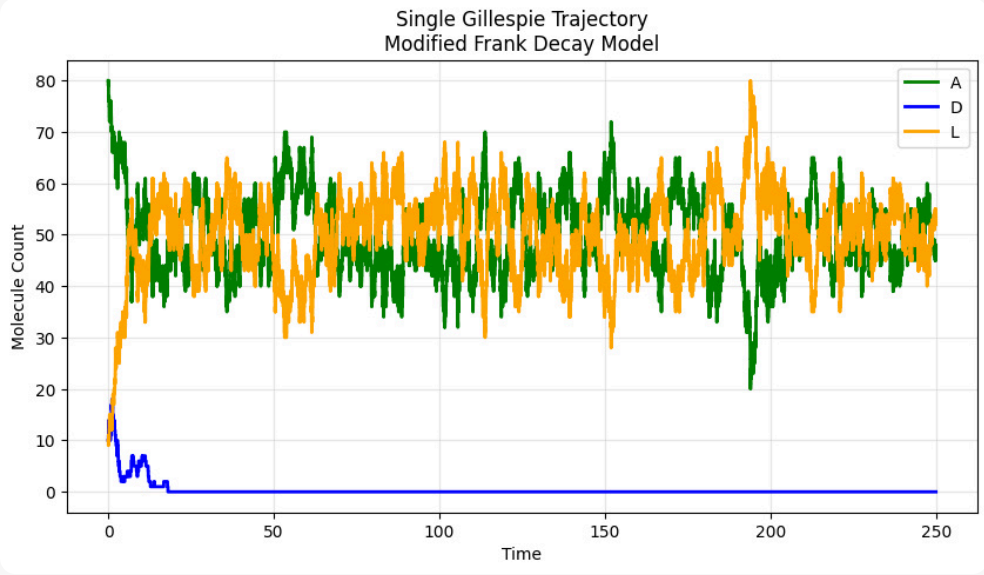
Gillespie Simulation:

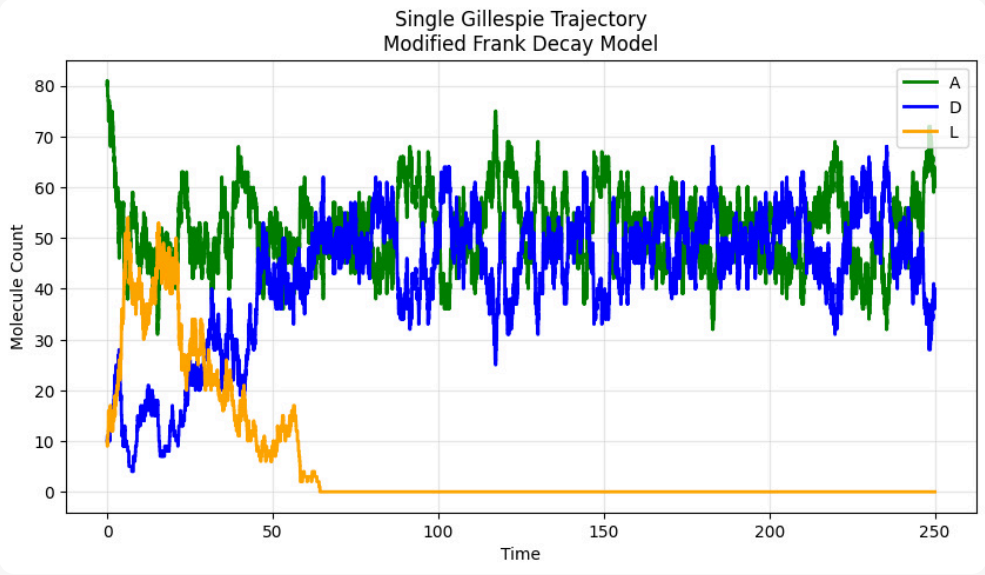
Gillespie simulation was performed over 1000 trajectories over the same assumptions that was taken for the deterministic simulation (A_0=0.8; $D_0 = L_0$=0.1 (starting conditions))



The ensemble trajectories do seem to be in agreement with the deterministic simulations however the individual runs seem as if theres more to uncover

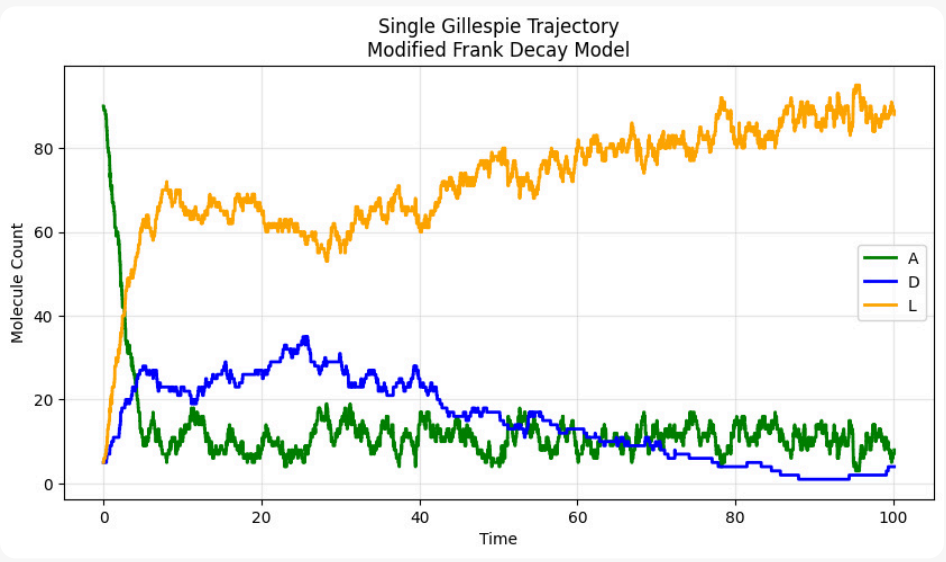
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These are two individual trajectories with the above conditions with no starting bias (complete racemic starting point). They yield in complete different results as it could appear that we have managed to break through to homochirality if the decay is much lesser than the rate of autocatalysis. However unlike the Frank's model what we find is that it cannot completely elimiate the chiral precursor A molecule as at one stage the decay and autocatalysis do seem to be competitive .

On further lowering the rate of decay what was found was that we could arrive at a position where we did manage to attain a state close to perfect homochirality however we could still not elimiate the chiral precurose A molecule.



This system had a similar construct except

k_a=1.0 and k_d=0.1 the decay is much lesser so it allows the initially dominant series to rise up, it has similar deterministic conditions to our last case where as here $D + L = 0.9$ as it approaches a stable equilibrium.

We did manage to get close however unline frank's model we could not arrive at a state where both of the non dominant enantiomer and the achiral precurosor were completely eliminated.

Conclusion:

The study did yield useful and informative results espeically at lower regime of decay rate values. However it still doesnt answer our question of what happens if we could reintroduce species which have completely decayed in the medium. The final model of our discussion would analyse that situation using reversible reactions.

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References

- **Frank, F. C. (1953).**

> Frank, F. C. *On spontaneous asymmetric synthesis*. Biochimica et Biophysica Acta, **11**, 459–463.

- **Jafarpour et al.**

> Jafarpour, F., Biancalani, T., & Goldenfeld, N. (2015). *Noise-induced mechanism for biological homochirality*. Physical Review Letters.

- **Soai Reaction**

> Soai, K., Shibata, T., Morioka, H., & Choji, K. (1995). *Asymmetric autocatalysis and amplification of enantiomeric excess*. Nature.

Software

> Python 3.13

Python was extensively used for all calculations and producing every image and statistic behind this presentation.

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Future Directions:

Going forward we will analyse the model that was newly proposed by **Jafarpour et al.**

Thank You!

